



PRODUCT BRIEF · HUMAN-STATE ASSURANCE · 2026

# Human-State Assurance for Mission Systems.

**Trusted data and AI still need a ready human decision-maker.**

Mission systems now move faster than the human decision loop. Defence platforms fuse sensors, AI, autonomy, command-and-control, cyber, and multi-domain operations — but the operator remains the final point of **interpretation, authorization, escalation, restraint, and recovery.**

## THE STAKES / WHY THIS MATTERS NOW

**Latency is not only in networks. It is also in cognition.**

## FORCE PRESERVATION GAP / UPSTREAM READINESS INDICATORS

**523**

**U.S. service-member deaths by suicide in CY2023**

DoD Annual Report on Suicide in the Military, CY2023.

U.S. Department of War

**17.5/day**

**Veteran suicides per day in 2023**

VA 2025 National Veteran Suicide Prevention Annual Report.

Mental Health VA

**6,398**

**Veteran deaths by suicide in 2023**

VA reporting, 2023.

VA News

## SLEEP-DEPRIVATION DECLINE / VERIFIED PERFORMANCE LOSS

**61%**

**marksmanship vigilance under sleep deprivation**

*fell from ~85% rested, after two nights of sustained operations*

McLellan et al. — simulated urban operations.

Aviat Space Environ Med

**44%**

**urban-operations vigilance under sleep deprivation**

*fell from ~78% rested, after two nights of sustained operations*

McLellan et al. — simulated urban operations.

Aviat Space Environ Med

**~25%**

**cognitive throughput decline**

*per successive 24 hours awake (total sleep deprivation)*

Thorne / Belenky, WRAIR — useful mental work, computer-based testing.

WRAIR

THE GAP · WHY CIQ IS DIFFERENT · 2026

# Presence Is Not Readiness.

*A present, cleared operator can still be cognitively degraded.*

## THE GAP / PRESENCE IS NOT READINESS

- A person can be **present, trained, cleared, and assigned** — yet still be cognitively degraded.
- Fatigue, overload, ambiguity, VUCA pressure, and recovery failure can slow reaction time, narrow attention, and increase cognitive latency.
- Failure often appears in the mission log only **after** the human state has already degraded.

## BASELINE-FIRST / WHY CIQ IS DIFFERENT

CIQ compares operators against **their own cognitive signature** — not generic population thresholds. Baseline modelling tracks normal patterns in reaction time, decision latency, attention stability, recovery state, HRV and behavioural variability — so deviations can be interpreted with context and confidence.

## THE NEW CATEGORY / HUMAN-STATE ASSURANCE

### WHAT IT IS

- ✓ A human-readiness intelligence layer
- ✓ Measures load, latency, attention, recovery, and burnout risk
- ✓ Supports earlier intervention before mission failure

### WHAT IT IS NOT

- ✗ Not wellness
- ✗ Not surveillance
- ✗ Not a lie detector or medical diagnosis



Modern defence systems already measure platforms, files, networks, sensors, and AI outputs. CIQ measures the one layer still weakly instrumented: **the human decision-maker under pressure.**

**VUCA** — Volatility, Uncertainty, Complexity & Ambiguity. Operating conditions that compress decision time and accelerate human-state degradation.

PRODUCT BRIEF · OPERATIONAL VALUE · 2026

# Operational Value for Defence Primes.

*How CIQ strengthens training, readiness and human-in-the-loop decision quality.*

## WHAT LEADERS SEE / THE INTELLIGENCE LAYER

|                |                  |                         |                        |
|----------------|------------------|-------------------------|------------------------|
| Cognitive Load | Decision Latency | Attention Stability     | Stress Response        |
| Recovery State | Burnout Risk     | Challenge-Skill Balance | Reversible Containment |

Aggregate readiness intelligence without exposing private individual profiles.

## HOW IT WORKS / THE LIFECYCLE

|                                                                          |                                                                |                                                                     |                                                                 |                                                                |
|--------------------------------------------------------------------------|----------------------------------------------------------------|---------------------------------------------------------------------|-----------------------------------------------------------------|----------------------------------------------------------------|
| <b>01</b><br><b>ASSESS</b><br>AI-led interview and structured assessment | <b>02</b><br><b>MEASURE</b><br>Signals mapped against baseline | <b>03</b><br><b>MODEL</b><br>Load, latency, overload risk, recovery | <b>04</b><br><b>ACT</b><br>Private reports and command insights | <b>05</b><br><b>VALIDATE</b><br>AAR and training-effectiveness |
|--------------------------------------------------------------------------|----------------------------------------------------------------|---------------------------------------------------------------------|-----------------------------------------------------------------|----------------------------------------------------------------|

## WHERE CIQ EARNS ITS KEEP / USE CASES

|                                             |                                             |                                               |                                                    |                                                        |                                                    |
|---------------------------------------------|---------------------------------------------|-----------------------------------------------|----------------------------------------------------|--------------------------------------------------------|----------------------------------------------------|
| <b>Strategic Wargaming</b><br>COA rehearsal | <b>Mission Readiness</b><br>Unit visibility | <b>Training Validation</b><br>Challenge-skill | <b>After-Action Review</b><br>Human-state sequence | <b>Operational Monitoring</b><br>Overload and recovery | <b>Human-in-the-Loop AI</b><br>Supervisory control |
|---------------------------------------------|---------------------------------------------|-----------------------------------------------|----------------------------------------------------|--------------------------------------------------------|----------------------------------------------------|

## PRIVACY-PRESERVING / COMMAND INTELLIGENCE

| INDIVIDUAL                                                                                                                                                                      | LEADER                                                                                                                                                                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>✓ Private readiness report</li><li>✓ Recovery and recalibration pathways</li><li>✓ No exposure of personal profile to superiors</li></ul> | <ul style="list-style-type: none"><li>✓ Aggregate unit-readiness insights</li><li>✓ Access-controlled dashboards</li><li>✓ Earlier intervention before mission failure</li></ul> |

## THE ASK / 30-DAY HUMAN-STATE ASSURANCE PILOT

**1 cohort → baseline → scenario → outcomes → after-action review**

Validate **decision latency, attention stability, cognitive load, recovery state, NASA-TLX**, instructor scoring and performance outcomes.

**Measure the layer still left to intuition.**

CONSCIENCEIQ.AI